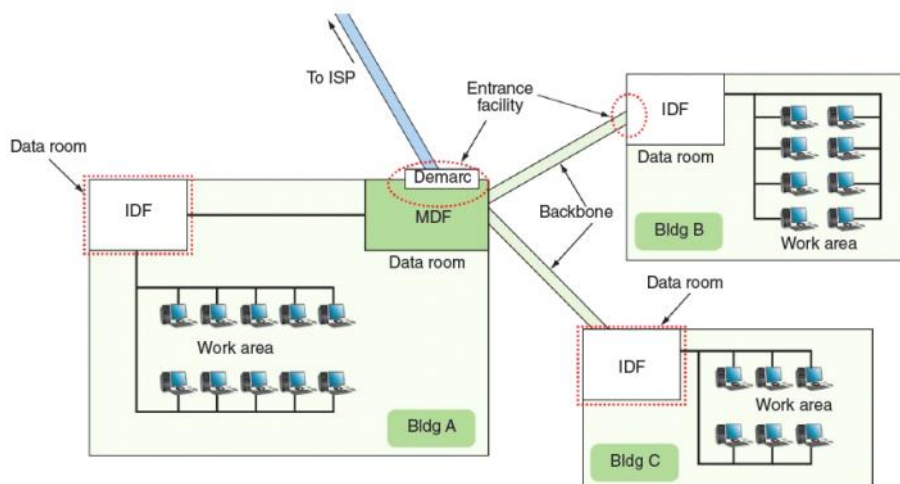


Network Equipment in Commercial Buildings

- TIA/EIA 568 is a wiring standard used in commercial buildings.
- It is also called structured cabling.
- It explains the best way to install cables for good performance and low maintenance.
- It works with any media type, technology, or speed.
- It uses a hierarchical design and star topology.



Components

- Entrance facilities
- MDF (main distribution frame)
- IDF (intermediate distribution frame)
- Horizontal wiring
- Backbone wiring
- Work area

Figure 2-1 TIA/EIA structured cabling in a campus network with three buildings

Entrance facilities:

This is where the **internet or network service enters the building from outside**. It includes the cables and equipment at the entry point.

MDF (Main Distribution Frame):

The MDF is the **main place in the building where all the network connections come together**. It connects the entrance facility to other parts of the building.

IDF (Intermediate Distribution Frame):

The IDF is **found in each floor or area of the building**. It connects the main distribution frame to local devices like computers or printers.

Horizontal wiring:

This is the **cable that runs from the IDF to the work area**. It connects wall ports to devices like computers.

Backbone wiring:

This is the **main cable that connects the MDF to each IDF**. It usually carries more data and is longer than horizontal cables.

Work area:

The **place where employees or users work**. It includes computers, phones etc.

!! components

Cable Management

- Don't leave too much of the cable uncovered before ending it.
- Don't bend cables too much.
- Test cables to check if they send data well.
- Tie cables gently, not tightly.
- Don't run cables across the floor; use covers if needed.
- Keep cables away from EMI (electromagnetic interference). Make sure cable sheath is plenum-rated.
- Follow grounding rules carefully.
- Leave extra cable slack.
- Use tools like cable trays or furniture grommets to organize cables.
- Use patch panels to keep lines neat and connected.



- Set cable standards for your organization.
- **Keep cable records** easy to find.
- Label every data jack or port, patch panel and connector.
- Use **color-coded tables for different purposes**.
- **Update** records when making changes.

Device Management

- Keeping device records up to date makes your job easier.
- It helps with fast and simple troubleshooting.
- **Name** and **label** devices clearly to track them easily.

Labeling and Naming Conventions



Tips for labeling;

- Use **color-coded cables** for dif. pruposes and tags to show each cable's purposes.
- **Label ports** and **jacks** clearly.
- Use **strong, sticky labels** made for plastic and metal.
- If labels don't fit, draw a diagram to show port use.
- Choose **clear** and **definer** names.
- **Follow standard naming rules**.

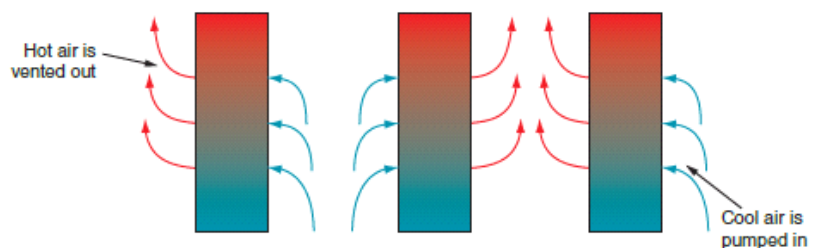
Rack Systems



- Rack systems **help save space** in equipment rooms by organizing network hardware efficiently.
- They also provide proper spacing, easy access, and good airflow for the devices.
- Most racks are 19 inches wide, but some are 23 inches.
- Racks can be on the wall, ceiling, or floor.
- Rack size is measured in rack units (RU or U).



Figure 2-11 Open two-post racks and enclosed four-post racks



- In a normal rack system, air usually moves from the **front to the back** to cool devices.
- A rack-monitoring system should be used to **alert** if the temperature becomes too high.

**! difference between nas and sans
(know meaning)**

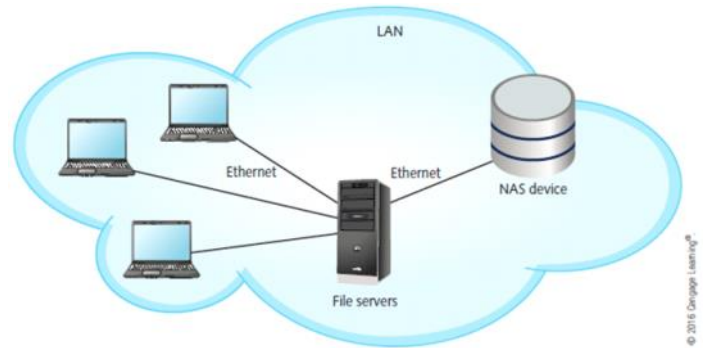


Figure 4-16 Network attached storage (NAS) on a LAN

NAS (Network Attached Storage)

- **Fault tolerance** refers to methods that allow data storage and operations to continue working even if one component fails.
- **NAS (Network Attached Storage)** is a **special storage system** that provides **centralized** and reliable data storage for a network:
 - It has its **own file system designed** for saving and sharing files efficiently.
 - It **reads** and **writes data** faster than many other server types.
 - It can be expanded easily without stopping the service.
-  Users do not connect directly to the NAS.
- Instead, clients connect to a file server, and the server handles communication with the NAS device.
- **NAS** is a system that stores files in a **central location** and allows everyone on the network to **access** them.

SANs (Storage Area Networks)

- A SAN is a network of storage **devices that talk to each other and other networks directly**.
 - It uses an architecture similar to mesh topology, which makes it very fault tolerant (reliable).
- SANs work with two Transport layer protocols:
 - Fibre Channel (FC)
 - iSCSI

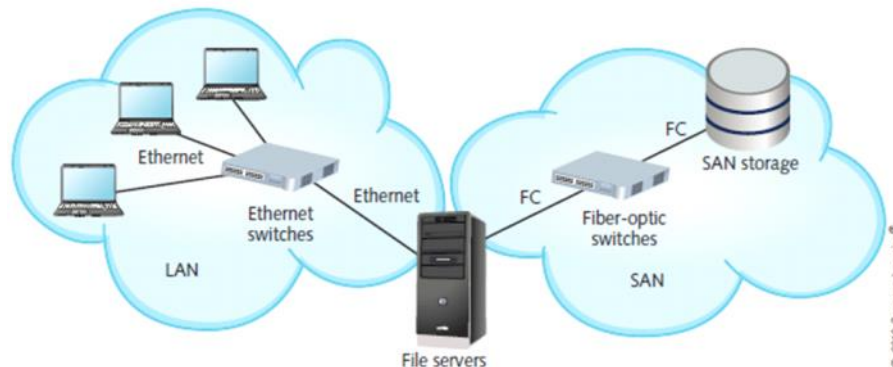


Figure 4-17 A storage area network (SAN) using Fibre Channel (FC) connected to a LAN

- A SAN can be placed in a different location from the LAN it supports.
- SANs are highly scalable and offer:
 - **A very high fault tolerance**
 - **Large storage capabilities**
 - **Fast data access**
- They are ideal for environments with large amounts of data that need to be available quickly.

Power Management

- Power surges can seriously damage sensitive computer devices and often cause network issues.
- To protect your systems, you should understand how electric circuits work and learn about the components that control electricity.

Power Flaws

There are several power issues that can damage your equipment:

- A **surge** is a sudden voltage spike caused by things like lightning or electrical faults.
- **Noise** is voltage fluctuation from other devices or electromagnetic interference.
- A **brownout** is a temporary voltage drop.
- A **blackout** is a total loss of power.

UPS (Uninterruptible Power Supply) (directly attached to the device and power supply)

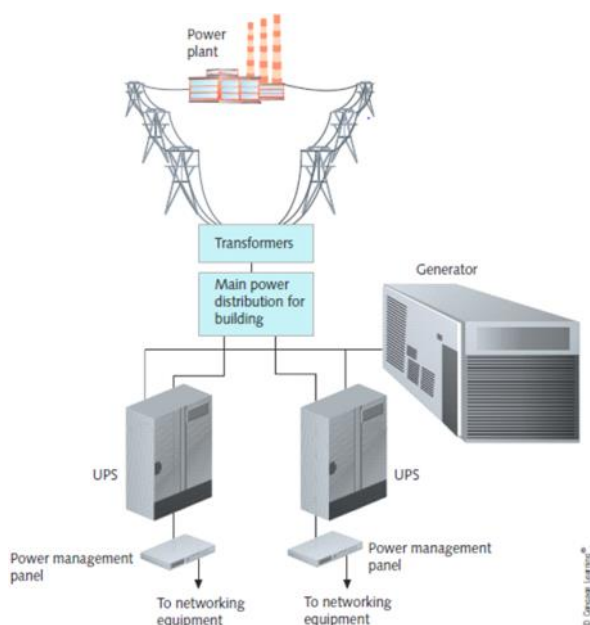
- A UPS is a battery-powered device that protects equipment from power loss or fluctuations.
- There are two types:
 - **Standby UPS**: Switches to battery when power is lost.
 - **Online UPS**: It always gives power from the battery and charges the battery using wall power.

When choosing a UPS consider these factors;

- power needs
- How long time to keep a device running
- Line conditioning
- Cost
 - It's also important to test the UPS with your own equipment to make the right choice.

Generators

- Generators provide backup power during a complete outage and can use various fuel types like diesel or gas.
- They are often used with **UPS** systems to make sure power stays clean and reliable.
- Organizations should check fuel levels and quality.



NICs and Ethernet

- Every network device like a computer or printer needs a NIC (Network Interface Card).
- On local networks, the NIC communicates using Ethernet standards.

Characteristics of NICs



- A NIC includes a **transceiver** to send and receive data over a network.
- It works at both the Physical and Data Link layers by:
 - **Sending signals** and managing data frames
 - **Reading** physical addresses
 - **Controlling which device** can send data
 - Handling tasks like **prioritization** and traffic filtering

NICs (Network Interface Cards) have different types based on:

- Connection type
- Maximum network transmission speed
- Number and type of ports
- Manufacturer
- Extra features like PoE+ ,buffering and traffic control
- How they connect to the motherboard and meet interface standards

A NIC connects to a computer's motherboard in three ways:

- Built into the motherboard
- Installed in an expansion slot
- Installed as an external device v

When installing NICs, you should:

- Install the required hardware and software
- Use peripheral or expansion card NICs
- Add multiple NICs to servers or powerful computers for better performance

SUMMARY

- TIA/EIA developed a standard called structured cabling to improve cable installation.
- Physical layer problems can happen due to bad cable ends, tight bends, damage, or EMI.
- Rack systems help save space and organize equipment in network rooms.
- NAS offers centralized and fault-tolerant data storage for networks.

Feature	NAS (Network Attached Storage)	SAN (Storage Area Network)
Definition	A dedicated file storage system connected to a network	A network of storage devices that connect directly to servers
Purpose	Centralized file sharing and storage	Fast access to large blocks of data
Connection	Devices connect indirectly through a file server	Devices connect directly to the storage
File System	Has its own file system	File system is managed by the client/server
Performance	Moderate speed	Very high speed

Fault Tolerance	Temel düzeyde	Gelişmiş, yüksek seviyede 
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